

MT 502: Recycling and Circular Economy

Course Code	MT 502
Course Name	Recycling and Circular Economy
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech/PG/Ph.D.
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction to unit operations:** Introduction to materials recycling and waste management, Categories of metallurgical and electronic waste, Key unit operations involved in pre-treatment of metallurgical and electronic waste, Application of pyrometallurgy, hydrometallurgy and electrometallurgy and refining for Material separation and enrichment of raw materials. [6 Hours]
- **Materials Cycle:** Materials and their cycle, Introduction to recycling sustainability, energy accounting minerals, Energy consumption in primary production, Management of wastes from primary production, Present and future availability and demand of minerals and metals, Techniques of sustainable extraction from secondary materials. [12 Hours]
- **Metallurgical Waste Recycling:** Aluminium industrial waste: dross, scrap, red mud, spent pot lining, salt slag. Copper industrial waste: smelter slags, raffinates, spent electrolytes, Zinc industrial waste: zinc ash, zinc dross, flue dust, and scraps, Iron and Steelmaking waste, scraps, ironmaking slag, steelmaking slag and wastewater. Recycling and reuse of blast furnace ironmaking slags, steel-making dust, and sludges. [8 Hours]
- **Electronic Waste Recycling:** Waste Printed Circuit Board, delamination and metal recovery, end-of-life batteries and recovery of valuable materials from waste electrodes, rare earth elements comprising electronic waste (LEDs, magnets, etc.), and environmental impacts of materials recycling. [6 Hours]
- **Introduction to Circular Economy:** From linear to circular economy, Cradle to Cradle concept, Eliminate waste and pollution, Circulate products and materials (at their highest value), Regenerate nature, Eliminate waste and pollution, Design changes, Focus on design to eliminate the concept of waste, Designers eliminating waste for a circular economy, Walter Stahel's The Performance Economy, William McDonough and Michael Braungart's Cradle to Cradle, Janine Benyus' Biomimicry, Industrial ecology, Regenerative design, Gunter Pauli's The Blue Economy, Systems thinking, Sustainable Development Goals (SDG) and linkages with Circular Economy model. [8 hours]

Laboratory Modules

1. NA

Text Books

1. P.C. Hayes, **Process principles in minerals and materials production-with a focus on metal production and recycling**, 2021.
2. L. Peter, **Materials for a Sustainable Future**, Royal Society of Chemistry Publishing, 2012.
3. J.A.S. Green, **Aluminum Recycling and Processing for Energy Conservation and Sustainability**, ASM International, 2007.
4. Ken Webster, **The Circular Economy: A Wealth of Flows**, 2nd Edition, 2017.
5. William McDonough, **Cradle to Cradle: Remaking the Way We Make Things**.
6. Andy Schmitz, **Sustainability, Innovation, and Entrepreneurship**. 1.0 Saylor Academy under a Creative Commons Attribution, Saylor Academy, 2012.
7. Amory Lovins, **A New Dynamic: Effective Business in a Circular Economy**.

References

1. S. Ramachandra Rao (Editor), **Resource Recovery and Recycling from Metallurgical Wastes**, Waste Management Series, 7, Elsevier.
2. Hugo Marcelo Veit and Andrea Moura Bernardes, **Electronic Waste Recycling Techniques**, Topics in Mining, Metallurgy and Materials Engineering, Springer International Publishing.
3. R. E. Heister and R. M. Harrison, **Electronic Waste Management**, Issues in Environmental Science and Technology, 27, RSC Publishing, Cambridge.